

Decision Tree Based NPC Behaviour in Video Game AI

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Abstract:

This project implements a Video Game AI system using a Decision Tree to simulate NPC behaviour. The system uses a Binary Tree structure in C where nodes represent decisions such as attack, patrol, or escape. It supports CRUD operations including adding, updating, deleting, and searching nodes. Dynamic memory allocation using malloc() and free() helps manage nodes efficiently.

Keywords:

Decision Tree, NPC Behaviour, Binary Tree, Game AI, C Programming

I. INTRODUCTION:

Artificial Intelligence (AI) enables Non-Player Characters (NPCs) in video games to make decisions based on game situations. Instead of fixed scripts, AI allows characters to react dynamically.

A Decision Tree is a common AI technique where decisions are organized in a tree structure. Each node represents an action and branches represent outcomes. This project uses a decision tree to model NPC behaviour.

II. PROBLEM STATEMENT:

NPC behaviour in many games is hardcoded, making it difficult to modify or extend. This leads to limited flexibility and complex updates.

This project solves the problem by implementing a dynamic decision tree system that supports CRUD operations.

III. OBJECTIVE

- Implement a decision tree for NPC behaviour
- Perform CRUD operations on nodes
- Use dynamic memory allocation in C

IV. DATA STRUCTURE USED

The system uses a Binary Tree.

Nodes represent actions such as:

- Attack Enemy
- Patrol Area
- Run Away

Each node stores an ID, decision text, and child pointers.

V. SYSTEM DESIGN

The system follows a tree-based design.

- Root node represents the main decision
- Child nodes represent possible actions
- Implemented as a menu-driven program

VI. ALGORITHM

Operations performed:

- Add Node
- Delete Node
- Update Node
- Search Node
- Display Tree

VII. IMPLEMENTATION

The program is developed using the C programming language.

Key features:

- Node representation using struct
- Dynamic memory allocation (malloc, free)
- Modular functions

VIII. RESULTS

The system successfully performs CRUD operations and correctly displays the tree structure for NPC decisions.

IX. CONCLUSION

The project shows how binary trees can be applied to AI decision making in games.

X. FUTURE SCOPE

- Graphical decision tree visualization
- Integration with game engines
- Advanced AI methods